

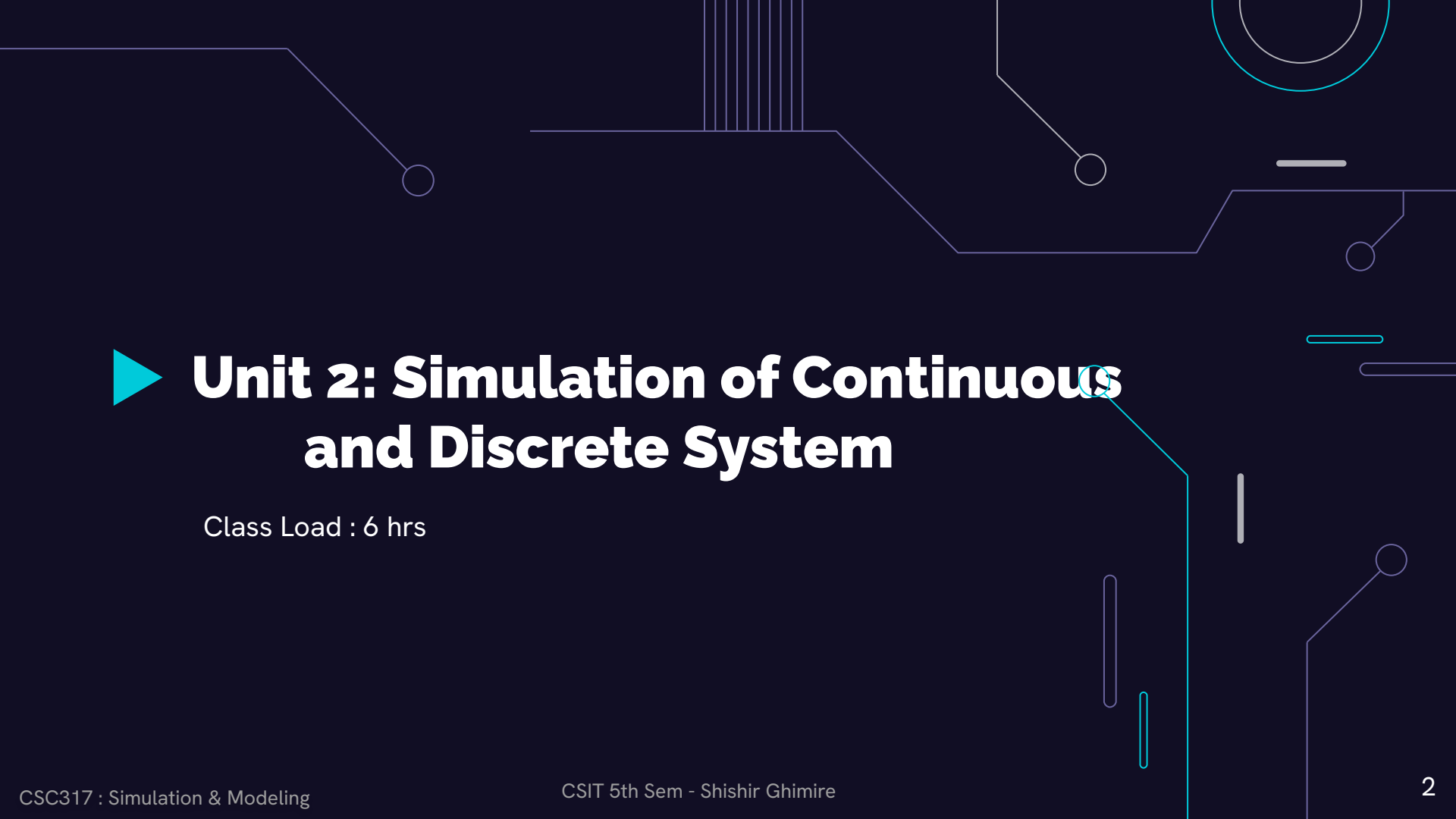


Simulation & Modeling

Course Code: CSC317
Year/ Semester: III/V

Compiled by Shishir Ghimire

Credit Hours: 3hrs



► Unit 2: Simulation of Continuous and Discrete System

Class Load : 6 hrs

► TABLE OF CONTENTS

Unit 2	Simulation of Continuous and Discrete System	7 Hours
Continuous System Models Analog Computer, Analog Methods, Hybrid Simulation, Digital-Analog Simulators, Feedback Systems	Concept, examples, use of differential equations for modelling continuous system	2 hours
Discrete Event Simulation, Representation of time, Simulation Clock and Time Management	Concept of discrete event simulation, time representation and management	1 hour
Arrival Processes - Poisson Processes, Non-stationary Poisson Processes, Batch Arrivals	Concept of arrival pattern, generation of arrival pattern using Poisson and Non-stationary Poisson with example, Introduction to batch arrival processes	2 hours
Models of Gathering statistics	Different statistics (like counts, summary measures, utilization, occupancy, distributions etc) that are needed to generate report and methods to gather such statistics	1 hour
Probability and Monte Carlo Simulation	Concept with an example	1 hour

► Differential and Partial differential equations:

Differential Equations

- ❖ A differential equation is an equation which contains one or more terms and the derivatives (instantaneous rate of change) of one variable (i.e., **dependent variable**) with respect to the other variable (i.e., **independent variable**).

$$dy/dx = f(x)$$

Here “x” is an independent variable and “y” is a dependent variable

For example, $dy/dx = 5x$

- ❖ An example of a linear differential equation with constant coefficients to describe the **wheel suspension system of an automobile** can be given as

$$M\ddot{x} + D\dot{x} + Kx = KF(t)$$

- ❖ The dependent variable x appears with its first and second order derivative $\dot{x}$ & $\ddot{x}$ and the term involving these quantities are multiplied by constant coefficient and added. The quantity $F(t)$ is an input to the system depending upon the independent variable t .

► Differential and Partial differential equations:

Partial Differential Equations

- ❖ A Partial Differential Equation commonly denoted as PDE is a differential equation containing **partial derivatives of the dependent variable (one or more)** with **more than one independent variable**.

- ❖ The simple PDE is given by;

$$\partial u / \partial x (x, y) = 0.$$

- ❖ It involves the derivative of the same dependent variable with respect to each of the independent variable.

$$1. \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$2. u_{xx} + u_{yy} = 0$$

► Differential and Partial differential equations:

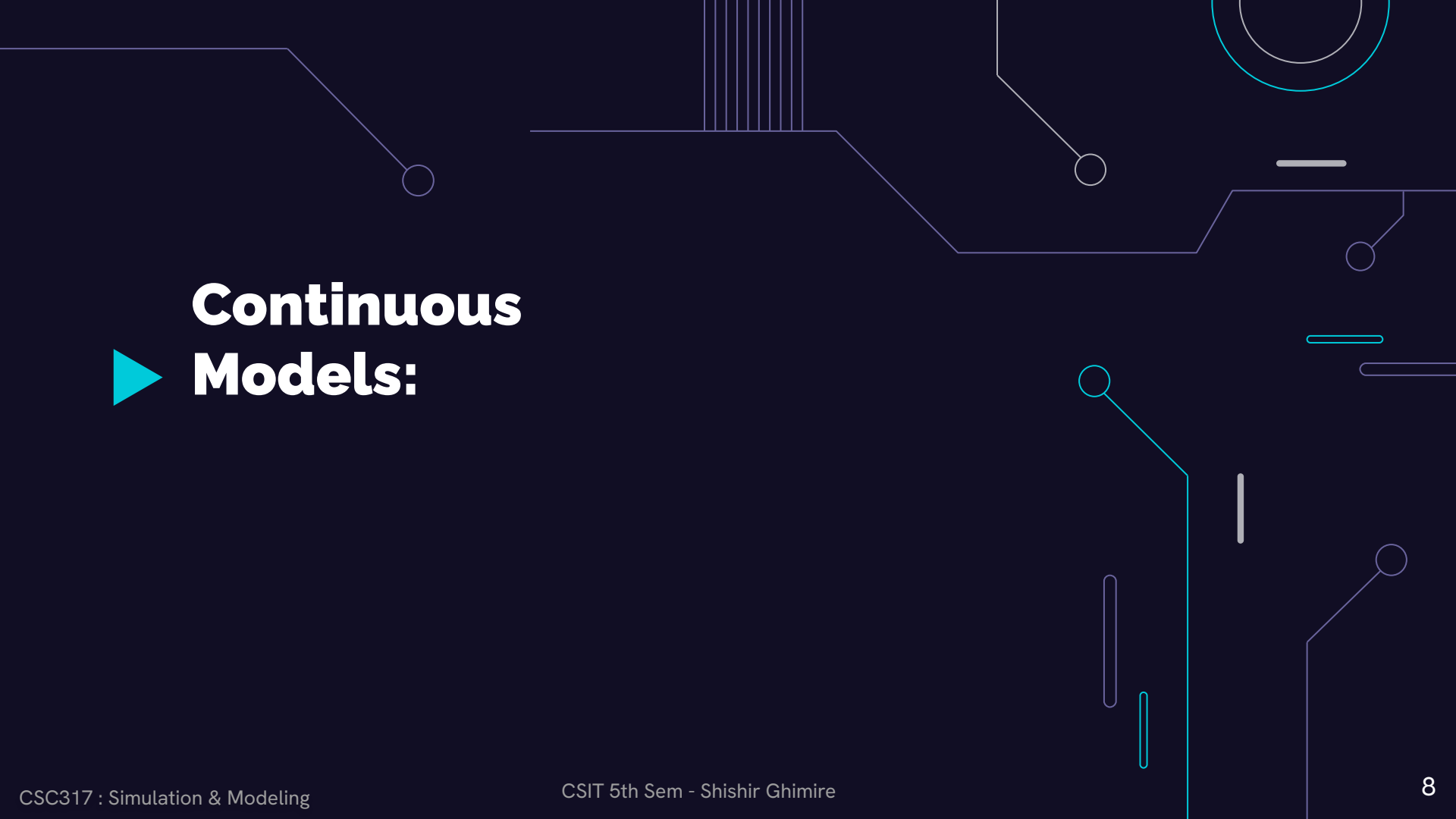
Linear Equations

- ❖ An equation in which the **maximum degree of a term** is **one** is called a linear equation. Or we can say that a linear equation that has **only one variable** is called a linear equation in one variable.
- ❖ A linear equation value when plotted on the graph **forms a straight line**.
- ❖ The general form of a linear equation is **$ax + b = c$** , where a, b, c are constants and x is a variable.
- ❖ Example
 - $x + 7 = 12$ is a linear equation

► Differential and Partial differential equations:

Non-Linear Equations

- ❖ Nonlinear equations are mathematical equations where the **degree of a variable term** is **two or greater**, or where variables are multiplied together.
- ❖ The nonlinear equation values when plotted on the **graph forms a curve**.
- ❖ The general form of a nonlinear equation is **$ax^2 + by^2 = c$** , where a, b, c are constants and x and y are variables. **Example:** $3x^2 + 2x = 7$ is a nonlinear equation.
- ❖ **Types of non-linear equation:**
 - Algebraic Equation
 - Polynomial Equation
 - Transcendental Equation



Continuous ▶ Models:

► Continuous System Simulation:

❖ Continuous System

- A continuous system is one in which the **state variable(s) change continuously over time.**
- **Analog or continuous signals** are used, allowing for smooth transitions and continuous changes.
- Key characteristics of continuous systems include:
 - Variables can take on any value within a certain range.
 - Behavior can be represented by continuous representations.
 - Analog representation (e.g., electrical circuits, fluid dynamics).
 - Continuous values (e.g., real numbers).

► Continuous System Simulation:

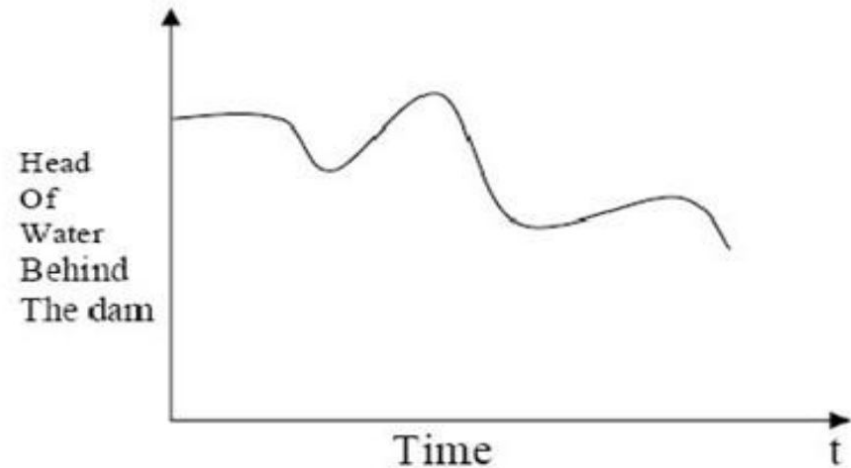
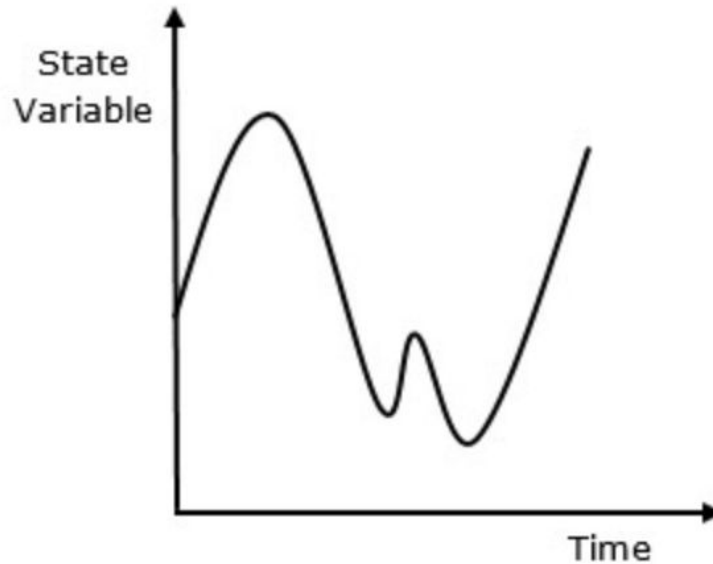
❖ Continuous System

➤ Example: Head of Water behind the Dam

- During winter seasons level of which water decreases gradually and during rainy season level of water increase gradually. The change in water level is continuous. The figure below shows the change of water level over time.

► Continuous System Simulation:

◆ Continuous System



► Continuous Models:

- ❖ **Continuous:** the state variables **change in a continuous way, and not suddenly** from one state to another (infinite number of states).
- ❖ Activities in a continuous system **completes smoothly without any delay**. i.e no queue of events, no sorting of time etc.
- ❖ A continuous simulation applies a **Continuous function using Real numbers** to represent a continuously changing system.
- ❖ Continuous simulations **consistently track system responses overtime** throughout the duration of a simulation task.
- ❖ Continuous simulation is great at **simulating flow processes**.

► Continuous Models:

- ❖ Continuous Simulation refers to a **computer model of a physical system** that continuously tracks system response according to a **set of equations** typically involving differential equations. **(linear equation to solve without use of simulation)**
- ❖ Continuous simulations are based on a set of differential equations.
- ❖ Continuous System Simulation describes systematically and methodically how mathematical models of dynamic systems, **usually described by sets of either ordinary or partial differential equations possibly coupled with algebraic equations, can be simulated on a digital computer.**

► Analog Computer:

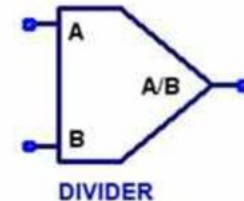
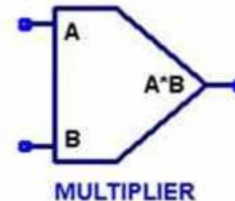
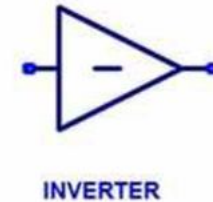
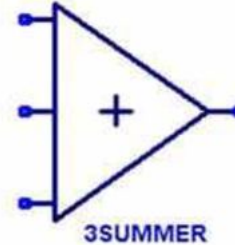
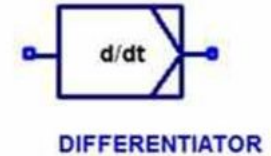
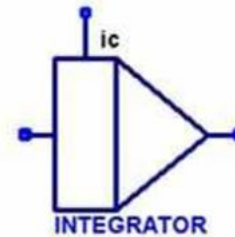
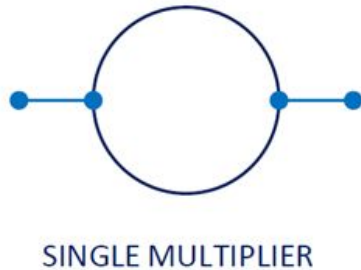
- ❖ An analog computer is a type of computing device that operates on continuous signals and physical quantities to perform calculations and simulations
- ❖ Before general availability of digital computers, there existed devices whose behaviour is equivalent to a mathematical operation, such as addition or integration.
- ❖ Putting together combination of such devices in a manner specified by a mathematical model of a system, allowed the system to be simulated.
- ❖ Analog computers can simulate continuous systems, such as the behavior of electrical circuits, fluid dynamics, mechanical systems, and other physical phenomena.
- ❖ By using electrical signals that continuously vary, analog computers can represent and predict the behavior of these systems over time.

► Analog Methods:

- ❖ Analog computers are those computers that are **unified** with devices like **adder (digital circuit) and integral** so as to simulate the **continuous mathematical model of the system**, which generates continuous outputs.
- ❖ Analog method of system simulation is for use of **analog computer** and **other analog devices** in the simulation of continuous system.
- ❖ The analog computation is sometimes called **differential analyser**. Electronics analog computers for simulation are based on the use of high gain DC amplifier called **operational amplifier (op amps)**.
- ❖ The general method to apply analog computers for the simulation of continuous system models involves **following components**:

► Analog Methods:

- ❖ The general method to apply analog computers for the simulation of continuous system models involves following **components**:



► Analog Methods for **Suspension Problem** :

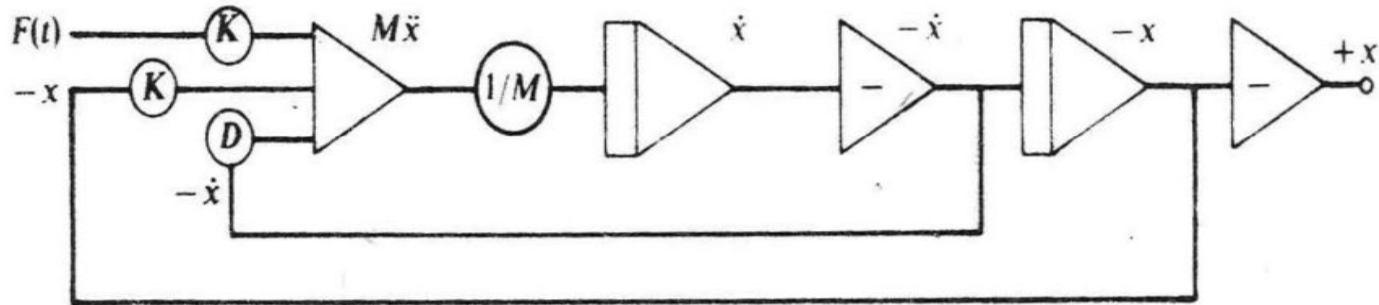
- ❖ The general method by which analog computers are applied can be demonstrated using **second order differential equation**. (**Example** : Automobile Suspension Problem)

$$M\ddot{x} + D\dot{x} + kx = kF(t)$$

- ❖ Solving the equation for the **highest order derivative** gives,

$$M\ddot{x} = kF(t) - D\dot{x} - kx$$

- ❖ The diagram to solve this problem in this manner is given in figure below as:



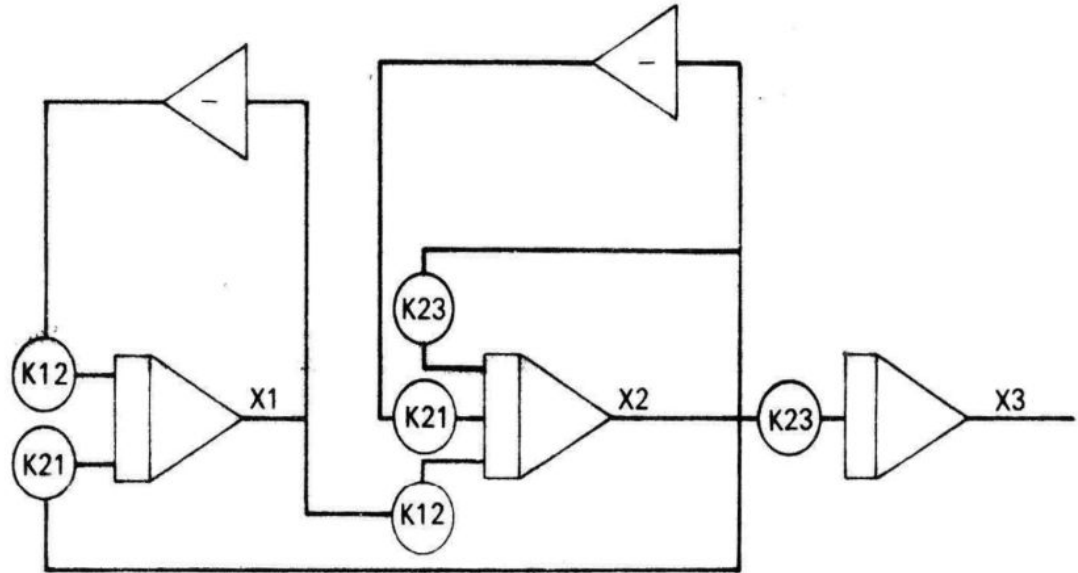
► Analog Methods ► **Suspension System** :

- ❖ Suppose a variable representing the **input** $F(t)$ is supplied, assume there exists variables representing $-x$, $-\dot{x}$ and $\ddot{x}$.
 - These three variables can be **scaled** and added to **produce** $M\ddot{x}$.
 - **Integrating** it with a scale factor $1/M$ produces $\dot{x}$.
 - Changing sign produces $-\dot{x}$, further integrating produces $-x$, a further sign inverter is included to produce $+x$ as **output**.

► Analog Methods ► Human Liver System :

❖ Q. Design analog computer of (Human Liver System)

- $\dot{x}_1 = -k_{12}x_1 + k_{21}x_2$
- $\dot{x}_2 = k_{12}x_1 - (k_{21} - k_{23})x_2$
- $\dot{x}_3 = k_{23}x_2$



Analog computer model of the liver

► Analog Methods ► Human Liver System :

- ❖ There are **three integrators**. Reading from left to right, they solve the equations for x_1 , x_2 & x_3 . **Interconnections** between the three integrators with sign changers where necessary provides inputs that define the differential coefficients of the three variables.
- ❖ First integrator, for example is solving the equation,

$$dx_1/dy = -k_{12}x_1 + k_{21}x_2$$

- ❖ The second integrator is solving the equation

$$dx_2/dy = k_{12}x_1 - (k_{21} - k_{23})x_2$$

- ❖ In this case, the variable x_2 is being used twice as an input to the integrator, so that the two coefficients k_{21} and k_{23} can be changed independently. The last integrator solves the equation

$$dx_3/dy = k_{23}x_2$$

► Hybrid Simulation:

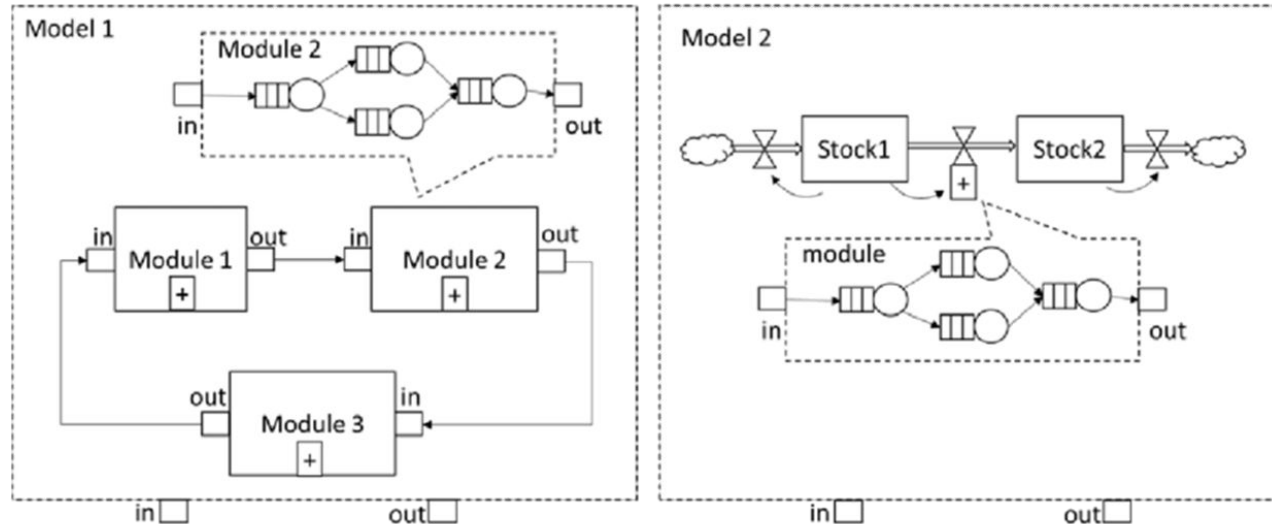
- ❖ For most studies, the model is clearly either of a **continuous or discrete** nature and that is the determining factor in deciding whether to use an **analog or digital computer for system simulation**. However, there are times when an analog and digital computers are **combined** to provide simulation.
- ❖ Currently the main application areas are healthcare, supply chain management and manufacturing, and the majority of published models combine **discrete-event simulation and system dynamics**.
- ❖ Here **one computer may be simulating the system** being studied while **other is providing a simulation of the environment** in which the system is to operate.
- ❖ It is also possible that the system being simulated is an **interconnection** of **continuous and discrete subsystems**, which can be modelled by an analog and digital computer **being linked together**.

► Hybrid Simulation:

- ❖ An **analog and digital computers are combined** to provide a hybrid simulation depending upon the application.
- ❖ It uses of an analog or a digital computer for system simulation **depends on the model** (continuous or discrete).
- ❖ The system being simulated is an interconnection of continuous and discrete sub system, which can best be modeled by an **analog and digital computer being linked together**.
- ❖ **High speed converters** are needed to transform signals from **one form** of representation **to** the **other**.
- ❖ Hybrid simulation is generally reserved for the case in which **functionally distinct** analog and digital computers are **linked together** for the purpose of simulation. **Example:** Use of digital elements added to the **operational amplifiers** of an analog computers.
- ❖ **Hybrid Simulation = Discrete Event Simulation (DES) + Continuous Simulation (CS)**

► Hybrid Simulation:

- ❖ Hybrid Simulation are the models that are simultaneously implemented on **both analog and digital computers**, or models that contain **both discrete and continuous variables**, or models that **combine** simulation with an analytical method such as optimization.



► Hybrid Simulation:

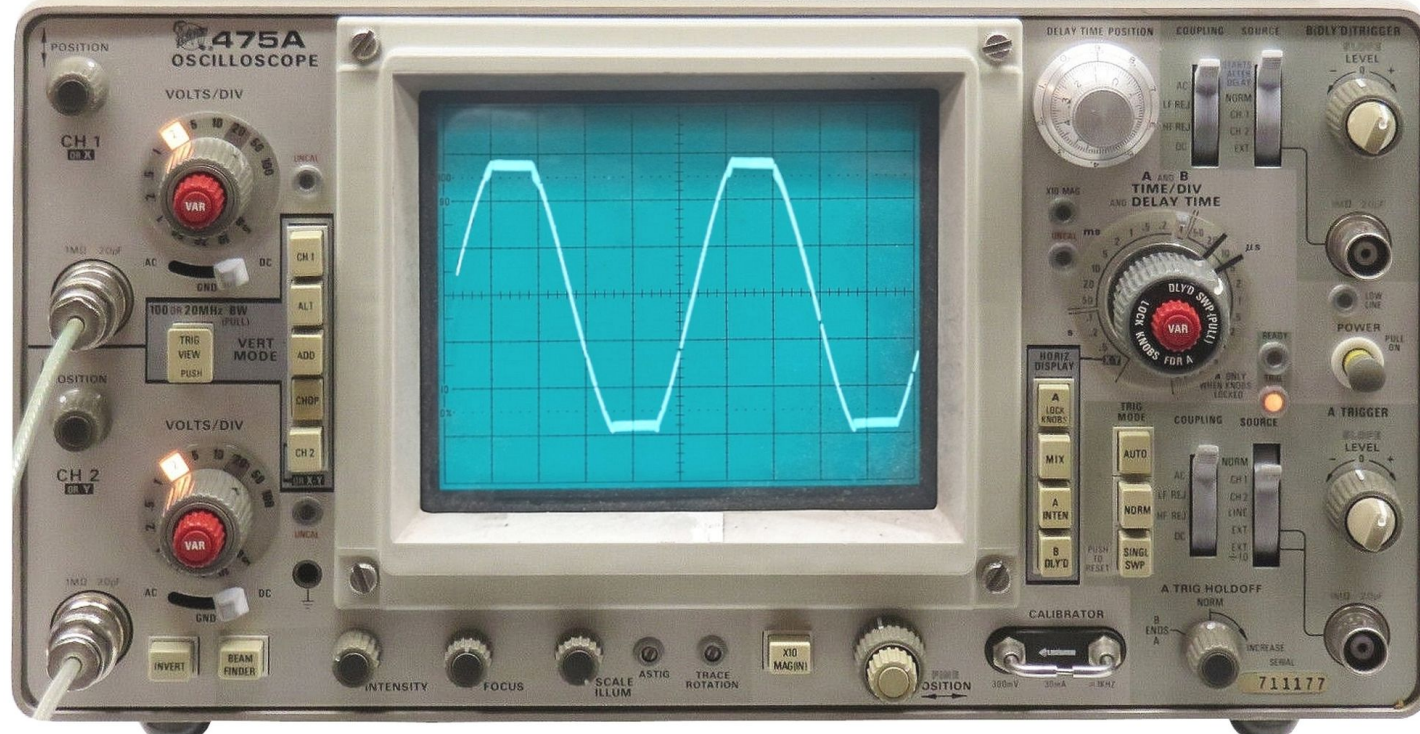
- ❖ They can be used to model systems **that are too large or too complex to be modeled with a single model.**
- ❖ They can be used to **combine the strengths** of different modeling approaches. For example, a **physical model** can be used to capture the dynamics of a system, while a **mathematical model** can be used to capture the uncertainty in the system.
- ❖ They can be used to validate numerical models. By **comparing the output** of a numerical model to the output of a physical model, it is possible to verify that the numerical model is accurate.
- ❖ **Example:**
 - A **physical model of a car** could be used to capture the **dynamics of the car's handling**, while a **mathematical model** could be used to capture the **uncertainty in the car's fuel consumption.**
 - A **physical model of a patient's heart** could be used to capture the **dynamics of the heart's beating**, while a **mathematical model** could be used to capture the **uncertainty in the patient's response to medication.**

► Analog vs Digital Computers:

Analog Computers

- ❖ A computer that uses a **continuous signal** to process is called an analog computer.
- ❖ The output of an analog computer in the form of **graphical or voltage signals**.
- ❖ Analog computers are **slower and less reliable**.
- ❖ Analog computers provide accurate computation results.
- ❖ Analog computers are used in specific fields such as mechanical engineering and medicine as they are **difficult to use**.
- ❖ **Example:** The best example of an analog device is **hands wrist watch** and **oscilloscope** (an instrument used to display and analyze the waveform of electronic signals) used by engineers for measurements.

► Analog Computer ► Oscilloscope :



► Analog vs Digital Computers:

Digital Computers:

- ❖ A computer that uses a **discrete signal** for its operation is called a digital computer.
- ❖ The output of digital devices is in the form of **binary numbers (0 and 1)**.
- ❖ Digital computers are **high-speed** devices that you can rely on.
- ❖ Digital computers **lose precision** due to the nature of discrete signals.
- ❖ These computers are designed to use for **general purposes**, thus can be used in all aspects of life.
- ❖ **Example:** The best example of digital devices are **PC, smartphones, calculators, a pedometer**, (which counts the exact number of steps while walking.)

► Digital – Analog Simulators

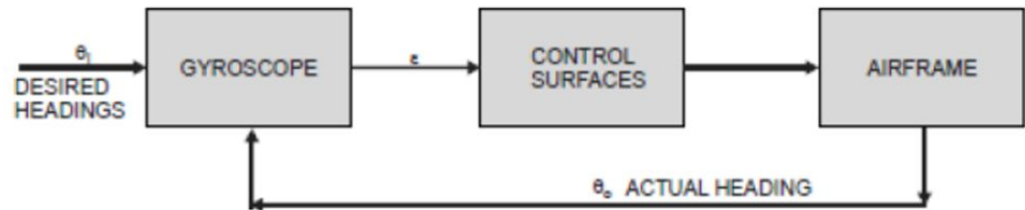
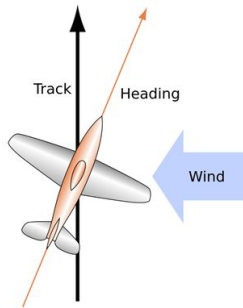
- ❖ To **avoid the disadvantages** of analog computers, many digital computer programming languages have been written to produce **digital analog simulators**.
- ❖ They allow a continuous model to be programmed on a **digital computer** in essentially the same way as **it is solved on an analog** computer.
- ❖ A **program** is written to **link** the **macro instructions** (addition, integration etc), in essentially same manner as operational amplifier are connected in the analog computer.
- ❖ Digital Analog Simulation is a **technique which makes a digital computer operate much like an analog** computer.

► Feedback System:

- ❖ A feedback system is **one that compares its output (to a desired input) and takes corrective action to force the output to follow the input.**
- ❖ The term feedback is used to describe the phenomenon (situation).
- ❖ A home heating system controlled by a **thermostat** (a device that automatically regulates temperature) is a simple example of a feedback system.
- ❖ **Example :**
 - The automatic heater whose **purpose is to heat a room**, and the **output** of the system can be measured as a room temperature. Depending upon whether the **temperature is below or above the thermostat setting**, the heater will be turned on or off, so that information is being feedback from the **output to input**. In this case there are only two states either the heater is on or off.

► Feedback System:

- ❖ **Example:** Continuous System Control Mechanism using feedback is the Aircraft System.
- ❖ Here the **input** is a **desired aircraft heading** and the **output** is the **actual heading**. The **gyroscope** of the autopilot is able to **detect the difference** between the two headings.
- ❖ A feedback is established by using the difference to operate the control surface. Since change of heading will then affect the signal being used to control the heading.
- ❖ The **difference between the desired heading and actual heading** is called the **error signal**, since it is a measure of the extent to which the system from the desired condition.



► Feedback System:

- ❖ Error Correction Mechanism
- ❖ $Y(t)$ – real output, $D(t)$ – desired output
- ❖ Let error at that instant be;

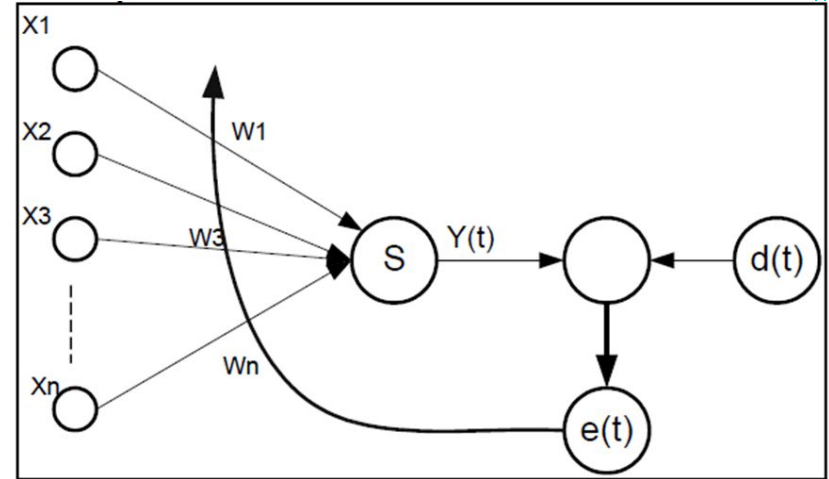
$$e(t) = D(t) - Y(t)$$

- ❖ $Y(t)$ is governed as;

$$\sum_{i=0}^n w_i(t).x_i(t)$$

- ❖ $e(t)$ is provided as feedback so that it adjusts the values of $w(t)$

The process is stopped if $Y(t) = D(t)$



► Feedback System:

1. Positive Feedback – control variable and output are **inversely proportional**.

- ❖ **Amplifies** the initial change, driving the system to completion or instability.
- ❖ The effect of positive feedback is to “**increase**” the systems gain, i.e, the overall gain with positive **feedback applied will be greater than the gain without feedback**.
- ❖ **Fruit Ripening**: Ethylene gas from ripe fruit causes nearby fruit to ripen faster.

2. Negative Feedback – control variable is **proportional with output**.

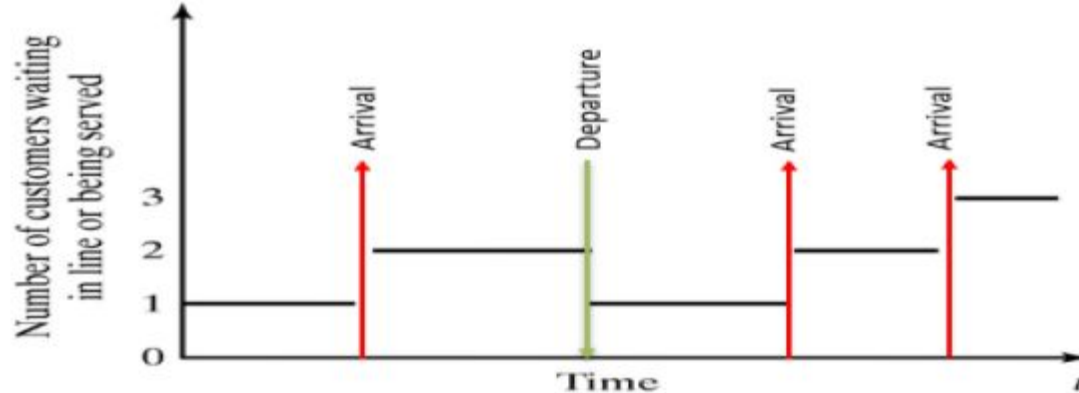
- ❖ **Reverses or counteracts** the initial change to maintain balance.
- ❖ The effect of negative feedback is to “**reduce**” the gain.
- ❖ **Blood Glucose**: High glucose → insulin release (lowers) resulting in Low glucose → glucagon release (raises).

► Discrete Event Simulation:

- ❖ A discrete-event simulation (DES) models the operation of a system as a discrete **sequence of events in time**. Each event occurs at a particular instant in time and marks a **change of state** in the system.
- ❖ Discrete event simulation (DES) is the process of codifying the behaviour of a complex system as **an ordered sequence of well-defined events**.
- ❖ Definition: DES is a process that models a system as a sequence of **distinct, instantaneous events, jumping time** between them, rather than tracking continuous change.
- ❖ **Example:** queues, manufacturing, or traffic

► Discrete Event Simulation:

- ❖ In a DES, system state evolves over a discrete sequence of events in time.
 - System State **changes only when** an event occurs
 - System state **does not change** in between the events.



Restaurant Example

► Discrete Event Simulation:

DES is **stochastic**, **dynamic**, and **discrete**.

- ❖ **Stochastic**: Some variables in the simulation model are **random**
 - **Inter-arrival times** and **service times** are random variables
- ❖ **Dynamic**: system **state evolves** over time
 - **Simulation clock**: Keep track of the current value of simulated time as the simulation proceeds.
 - A **mechanism to advance** simulated time from one value to another
- ❖ **Discrete-Event**: changes in system state occur at **discrete time instances**.
 - The state variables change instantaneously at separate points in time
 - The system can change at only a countable number of points in time.
 - These points in time are the ones at which an event occurs.

► Discrete Event Simulation ► Components :

To build or understand a DES model, we must understand the following components:

- ❖ **Attributes:** Properties of a given entity.
- ❖ **Event Notice:** A record of an event to occur at the current or some future time, along with any associated data necessary to execute the event
- ❖ **Event List (or FEL):** A list of event notices for future events, ordered by time of occurrence; also known as the future event list (FEL)
- ❖ **Delay (conditional wait):** A duration of time of unspecified indefinite length which is not known until it ends.
- ❖ **Activity (unconditional wait):** A duration of time of specified length which is known when it begins.

Note:

- Completion of an **activity** is an event (primary event).
- Completion of a **delay** is conditional/secondary event. Such events are not represented by event notices **nor do** they appear on the FEL

► Discrete Event Simulation:

- ❖ An effective DES process **must include**, at a minimum, the **following characteristics**:
 - Predetermined **starting and ending points**, which can be discrete events or instants in time.
 - A method of **keeping track of the time** that has elapsed since the process began.
 - **A list of discrete events** that have occurred since the process began.
 - **A list of discrete events pending** or expected (if such events are known) until the process is expected to end.
 - A graphical, statistical, or tabular record of **the function for which DES is currently engaged**.

► Discrete Event Simulation:

Simulation of Document Processing (Printing):

Document Number (i)	Start Time (tb)	Work Time (tw)	Finish Time (tf)	Cumul. Time (tc)	Break Flag (F)	No. of jobs (N)
1	0	45	45	45	0	7
2	45	16	61	61	1	6
3	66	5	71	5	0	5
4	71	29	100	34	0	4
5	100	33	133	67	1	3
6	138	25	163	25	0	2
7	163	21	184	46	0	1

► Time Representation:

❖ Clock Time:

- Clock time is the **real-world time** that records how many **units of simulated time have passed since the beginning of the simulation**.
- It is measured by the **computer's clock** and is the same for all users of the simulation. It is the time that would be measured by a stopwatch or a clock in the real world.

❖ Simulation Time:

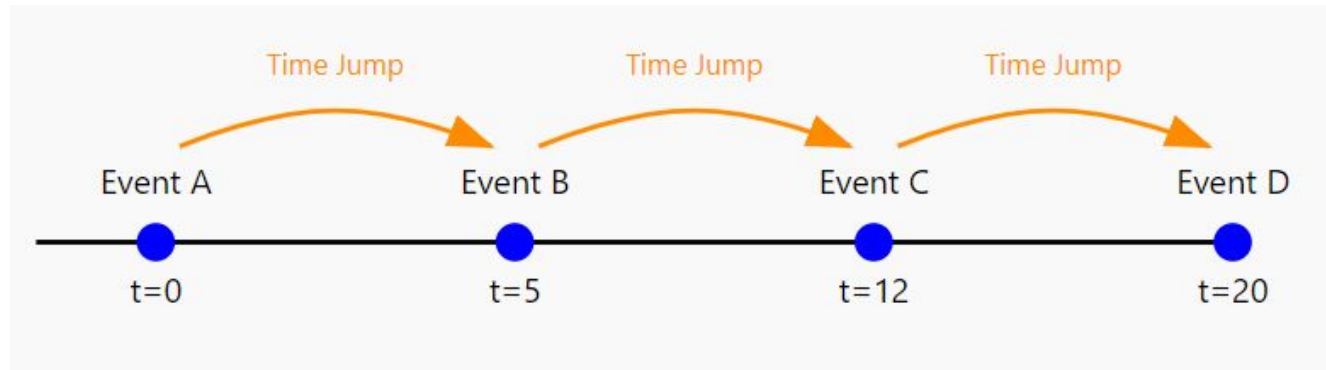
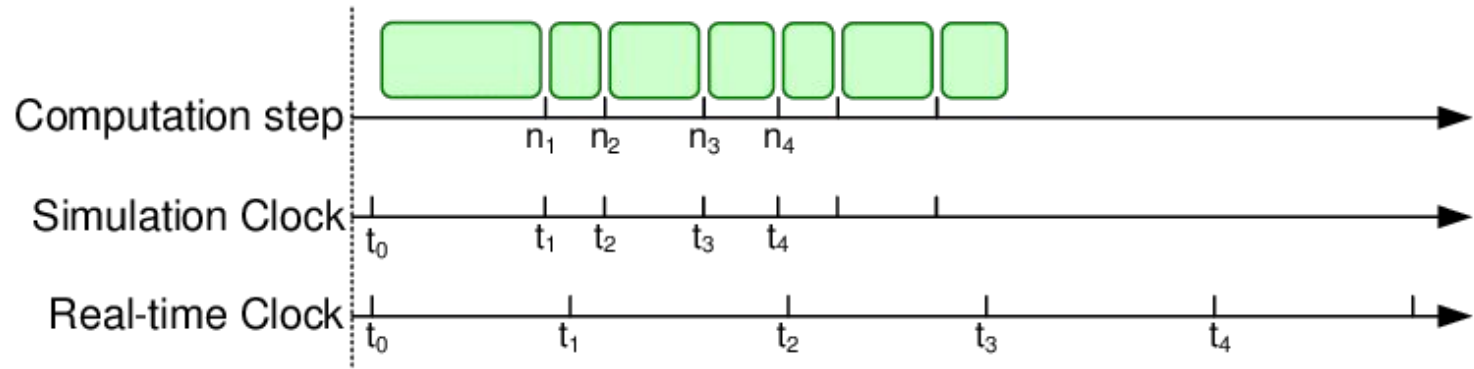
- Simulation Time is the **time that is simulated in the model**.
- It is not necessarily the same as clock time, and can be **faster or slower than clock time** depending on the speed of the computer and the complexity of the model.
- It is an **virtual time** used within the simulation to model the passage of time in the simulated environment.

E.g.: Let's say you are running a simulation of a queueing system. **The clock time might be 10 minutes**, but the **simulation time might only be 5 minutes** if the simulation is running on a fast computer. This is because the simulation only has to simulate the events that occur in the queueing system, and it can skip over the time when there are no events.

► Time Representation ► Clock vs Simulated Time :

Feature	Clock Time (Wall-Clock)	Simulated Time (Virtual Time)
Definition	The actual time elapsed in the physical world (e.g., measured by a wristwatch).	The time value inside the simulation model representing the passage of time in the modeled system.
Progression	Continuous and fixed; 1 second always takes 1 second of real life.	Discrete or "jumping"; the clock "hops" from one event to the next, skipping empty intervals.
Speed	Determined by the Earth's rotation and physical laws.	Determined by computer processing power and model complexity; can be much faster or slower than real time.
Purpose	Used to measure computational cost, efficiency, and real-world deadlines.	Used to ensure logical causality and chronological order of events within the model.

► Time Representation:



► Time Advance Mechanism:

Two basic methods exist for updating clock time.

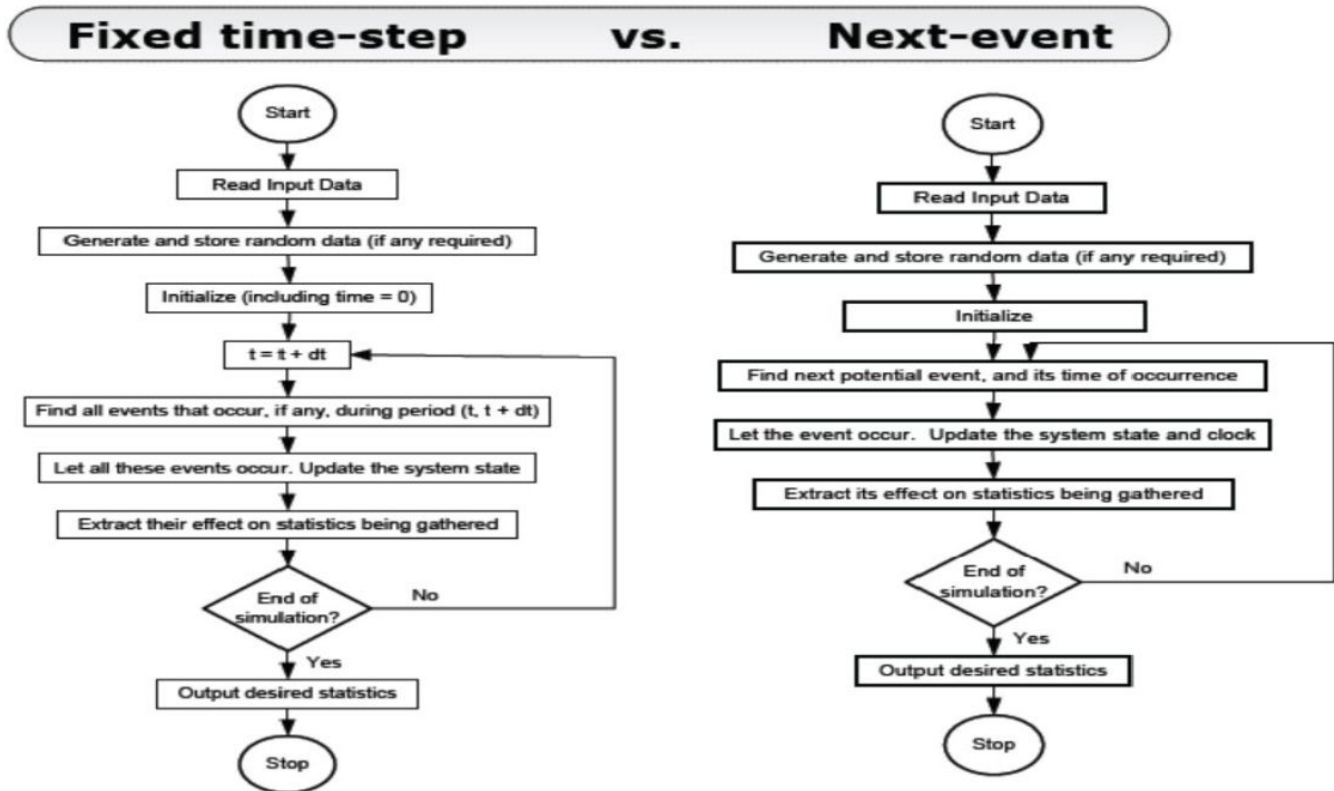
❖ Event oriented (Next Event Model):

- This method is to **advance the clock** to the time at which the **next event** is due to occur.
- **Discrete system** is carried out by event oriented method.

❖ Interval oriented(Fixed Time Step Model) :

- This method is to **advance the clock** by **small (usually uniform) intervals of time** and determine at each interval whether an event is due to occur at that time.
- **Continuous system** is carried out by using interval oriented method.

► Time Advance Mechanism ► (Flowchart) :



► Discrete Event Simulation Model:

The typical output expected from a discrete event model is:

- ❖ Utilization of resources
- ❖ Time spent in the system or its part by an agent
- ❖ Waiting times
- ❖ Queue lengths
- ❖ **System throughput**
 - Throughput is the **amount of a product or service that a company can produce and deliver to a client within a specified period** of time
- ❖ **Bottlenecks**
 - A bottleneck is a **situation that stops a process or activity from progressing**. when a machine is not efficient enough and as a result has a long queue.
- ❖ Cost of the agent processing and its structure.

► Gathering statistics:

- ❖ Most simulation programming system include a **report generator** to print out **statistics gathered during the run**. The exact statistics required from a model depend upon the study being performed, but there are certain commonly required statistics which are usually included in the output:
 - **Counts** - no. of entities of particular type or **no. of times some event occurred**
 - **Summary measures** - extreme values, mean values, standard deviations
 - **Utilization** - fraction (or percentage) of time some entity is engaged
 - **Occupancy** - fraction (or percentage) of a group of entities in use on the average
 - **Distributions** - queue lengths, waiting times etc.
 - **Transit times** - time taken for an entity to move from one part of the system to some other part

► Arrival Processes:

- ❖ How customers arrive e.g. singly or in groups?
- ❖ How the arrivals are distributed in time (e.g. what is the probability distribution of time between successive arrivals (the interarrival time distribution))
- ❖ Whether there is a finite population of customers or (effectively) an infinite number.

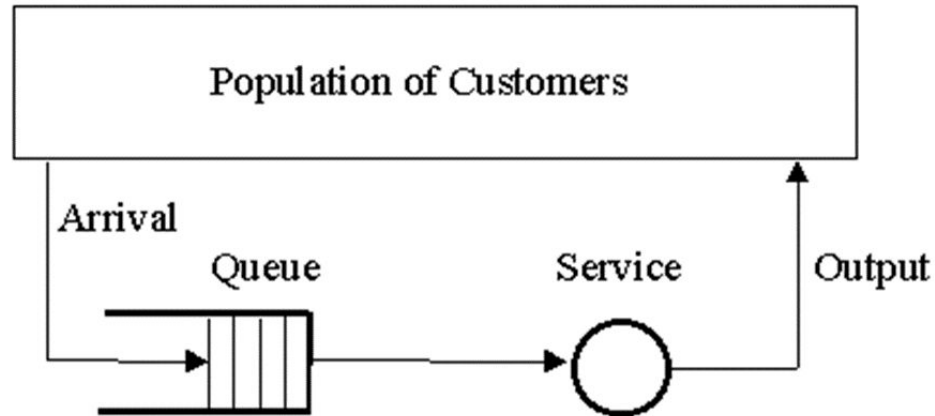
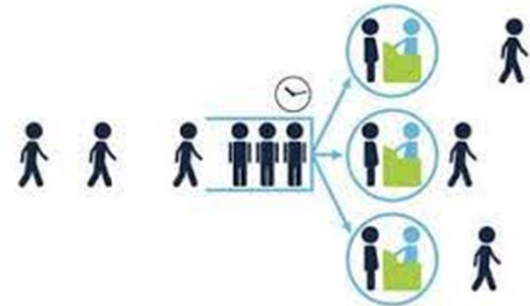
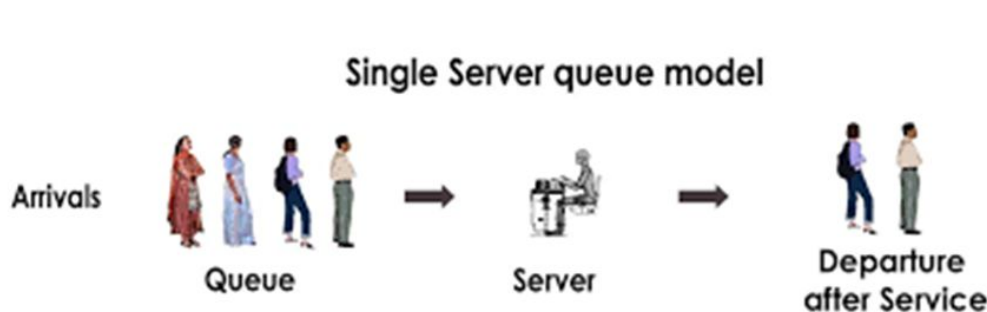


Figure 1

► Arrival Processes:

❖ Service mechanism:

- A **description of the resources** needed for service to begin
- How **long the service will take** (the service time distribution)
- The **number of servers** available
- Whether the **servers are in series** (each server has a separate queue) **or in parallel** (one queue for all servers)
- Whether **preemption** is allowed (a server can stop processing a customer to deal with another "emergency" customer)



► Arrival Processes:

Queue characteristics:

- ❖ How, from the set of customers waiting for service, do we choose the one to be served next
 - FIFO (first-in first-out) also known as FCFS (first-come first served)
 - LIFO (last-in first-out)
 - Service in Random Order
 - Shortest Processing Time First
 - Priority

Remember:

- ❖ **balking** (customers deciding not to join the queue if it is too long)
- ❖ **reneging** (customers leave the queue if they have waited too long for service)
- ❖ **jockeying** (customers switch between queues if they think they will get served faster by so doing) a queue of finite capacity or (effectively) of infinite capacity

► Arrival Processes:

- ❖ The simulation time **starts** when **first customer arrives** to the system
- ❖ The **arrival time** is random and hence **inter-arrival time** is also random

Customer	Inter-arrival time	Arrival time on clock
1	-	0
2	2	2
3	4	6
4	1	7
5	2	9
6	6	15

► Arrival Processes:

- ❖ If arrivals vary stochastically(randomly), it is necessary to define the **probability function** of the inter-arrival times.
- ❖ Two or more arrivals may be simultaneous.

❖ Notation:

- T_a - Mean inter-arrival time
- " λ " - Mean arrival time
- They are related as; " λ "= $1/T_a$

► Arrival Processes:

❖ Example - office

- five days a week
- Eight-hour day
- 800 calls a week

Calculate *mean arrival rate* and *mean inter-arrival time of calls*

❖ Model the office using time scale of minutes

- 1 week, 8 hours day = $8 \times 5 = 40$ working hours.
- Therefore inter-arrival time = $(40 \times 60) / 800$ minutes
= 3 minutes
- i.e. **arrival rate** = $1/3 = 0.333$ calls per minute

► Arrival Processes:

◆ Types of Arrival Patterns:

1. Poisson Process
2. Non Stationary Poisson Process

► Poisson Distribution:

- ❖ The Poisson distribution is a **probability distribution** that models the probability of a given **number of events occurring within a fixed interval of time or space**, provided these events happen:
 - Independently of each other, and
 - At a constant average rate (λ , also called the mean rate).
- ❖ A Poisson Distribution variable N with mean rate $\lambda > 0$ has a probability mass function (PMF) given by:

$$f(x) = P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots, \lambda > 0$$

Where:

λ is the **mean rate** and must be positive.

► Poisson Process:

- ❖ A Poisson Process is a **stochastic (random) process** used to model the occurrence and evolution of events over time (or space), when those events happen independently and at a constant average rate.
- ❖ A Poisson Process meets the **following criteria**:
 1. **Events are independent of each other**: The occurrence of one event does not affect the probability another event will occur.
 2. The **average rate (events per time period) is constant**.
 3. Two **events cannot occur at the same time**. (Customers arrive one at a time)
- ❖ **Examples**:
 - Common examples of Poisson processes are **customers calling a help center, visitors to a website, radioactive decay in atoms**, photons arriving at a space telescope, and **movements in a stock price**. Poisson processes are generally associated with time.

► Poisson Process ► Assumptions :

❖ Properties / Assumptions of Poisson Process

1. **Independent increments:** The number of events occurring in non-overlapping time intervals are **independent**.
2. **Stationary increments:** The probability of events depends only on the **length of the time interval**, not on its position.
3. **Single event occurrence:** In a very small interval Δt ,
 - Probability of one event $\approx \lambda \Delta t$
 - Probability of more than one event ≈ 0 [i.e Two **events cannot occur at the same time**]
4. **Initial condition:** $N(0)=0$

► Poisson Process:

- ❖ A Poisson Process is a model for a series of discrete event where the **average time between events is known**, but the **exact timing of events is random**.
- ❖ The Poisson process is one of the most widely-used **counting processes**. It is usually used in scenarios where we are **counting the occurrences of certain events** that appear to happen **at a certain rate**, but **completely at random** (without a certain structure).
- ❖ The arrival of an event is **independent** of the event.
- ❖ The Poisson Process can be used to **model the number of occurrences** of events.
- ❖ The **number of arrivals per unit time** is a random variable with a Poisson's distribution.

► Poisson Process:

- ❖ A Poisson process **helps to predict the probability of certain events happening when you know how often the event has occurred.** It gives us the probability of a given number of events happening in a **fixed interval of time.**

$$P(X = x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots$$

- Where λ is the average number of arrivals per unit time, and
 - x is the number of customers or occurrence per unit time.
 - e is Euler's number (constant).
- ❖ This **pattern of arrival** is called Poisson's arrival pattern or Poisson distribution.

► Poisson Process:

$$P(X = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

Where

λ = mean arrival rate per time unit.

t = the length of the interval.

$e = 2.7182818$ (the base of the natural logarithm).

$k! = k (k-1) (k-2) (k-3) \dots (3) (2) (1)$.

► Poisson Process vs Distribution:

Aspect	Poisson Distribution	Poisson Process
Meaning	Describes the probability behavior of a random variable	Describes the evolution of random events over time or space
Focus	Outcome at a point or interval	Sequence of events across time
Nature	Static	Dynamic
Output	Probability of a value	Number of events as time progresses
Mathematical Object	Random variable (X)	Stochastic process ($N(t), t > 0$)
Example	Number of phone calls received in 1 hour.	Customer arrivals at a bank over time.

► Poisson Process:

Example-1: The average number of major storms in your city is 2 per year. What is the probability that exactly 3 storms will hit your city next year?

- **Step 1:** Given
 - Average number of storms per year $\lambda=2$
 - Number of storms required $x=3$
 - $e = 2.71828$ (e is Euler's number, a constant)
- **Step 2:** By Poisson's Process Formula,

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$



$$\begin{aligned} P(X = 3) &= \frac{e^{-2} \cdot 2^3}{3!} = \frac{e^{-2} \cdot 8}{6} \\ P(X = 3) &\approx \frac{0.1353 \cdot 8}{6} \approx 0.1804 \end{aligned}$$

- The probability of exactly 3 major storms happening next year is 0.1804, or 18.04%

► Poisson Process:

Example-2: On a particular river, overflow floods occur once every 100 years on average. Calculate the probability of $k = 0, 1, 2, 3, 4, 5$, or 6 overflow floods in a 100-year interval, assuming the Poisson model is appropriate.

❖ Solution

- Average number of overflow floods per 100 years, $\lambda=1$
- Time interval = 100 years
- Number of floods required: $k=0,1,2,3,4,5,6$
- $e=2.71828$ (Euler's number)

$$P(k \text{ overflow floods in 100 years}) = \frac{\lambda^k e^{-\lambda}}{k!} = \frac{1^k e^{-1}}{k!}$$

$$P(k = 0 \text{ overflow floods in 100 years}) = \frac{1^0 e^{-1}}{0!} = \frac{e^{-1}}{1} \approx 0.368$$

For $k = 0$

$$P(0) = \frac{e^{-1}}{0!} = 0.3679$$

For $k = 1$

$$P(1) = \frac{e^{-1}}{1!} = 0.3679$$

For $k = 2$

$$P(2) = \frac{e^{-1}}{2!} = \frac{0.3679}{2} = 0.1839$$

For $k = 3$

$$P(3) = \frac{e^{-1}}{3!} = \frac{0.3679}{6} = 0.0613$$

For $k = 4$

$$P(4) = \frac{e^{-1}}{4!} = \frac{0.3679}{24} = 0.0153$$

For $k = 5$

$$P(5) = \frac{e^{-1}}{5!} = \frac{0.3679}{120} = 0.0031$$

For $k = 6$

$$P(6) = \frac{e^{-1}}{6!} = \frac{0.3679}{720} = 0.0005$$

► Poisson Process:

Example-3: Customers arrive at a shop as a Poisson process with rate 20 per hour. In a half-hour period, calculate the probability that 4 customers arrive in the first 15 minutes and 6 in the next 15 minutes.

Step 1: Given

- Arrival rate $\lambda = 20$ customers per hour
- Time interval = 15 minutes = $\frac{15}{60} = 0.25$ hour
- Mean number of arrivals in 15 minutes:

$$\lambda t = 20 \times 0.25 = 5$$

- Required:
 - First 15 minutes: $x_1 = 4$
 - Next 15 minutes: $x_2 = 6$
- $e = 2.71828$ (Euler's number)

❖ **Step 2:** For a Poisson process, numbers of arrivals in disjoint intervals are independent.

- So, $P(4 \text{ in first 15 min and } 6 \text{ in next 15 min}) = P(N_1 = 4) \times P(N_2 = 6)$
- So, the probability that 4 customers arrive in the first 15 minutes and 6 in the next 15 minutes is approximately 2.57%.

Poisson Formula

$$P(X = x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

Step 1: Probability of 4 customers in first 15 minutes

$$P(N_1 = 4) = \frac{e^{-5} 5^4}{4!} = \frac{e^{-5} \cdot 625}{24}$$

Step 2: Probability of 6 customers in next 15 minutes

$$P(N_2 = 6) = \frac{e^{-5} 5^6}{6!} = \frac{e^{-5} \cdot 15625}{720}$$

Step 3: Multiply (independence)

$$P = \left(\frac{e^{-5} 5^4}{4!} \right) \left(\frac{e^{-5} 5^6}{6!} \right) = \frac{e^{-10} \cdot 5^{10}}{4! 6!}$$

Numerical Value

$$P \approx 0.1755 \times 0.1462 \approx 0.0257$$

► Poisson Process ► Example-4 :

The number of customers arriving at a grocery store can be modeled by a Poisson process with intensity $\lambda = 10$ customers per hour.

1. Find the probability that there are 2 customers between 10:00 and 10:20.
2. Find the probability that there are 3 customers between 10:00 and 10:20 and 7 customers between 10:20 and 11.

Given

Poisson process with rate

$$\lambda = 10 \text{ customers/hour}$$

1 Probability of exactly 2 customers (10:00–10:20)

Time:

$$t = \frac{20}{60} = \frac{1}{3} \text{ hour}$$

Mean:

$$\mu = \lambda t = 10 \times \frac{1}{3} = \frac{10}{3}$$
$$P(N = 2) = e^{-10/3} \frac{(10/3)^2}{2!} \approx 0.224$$

2 Probability of 3 customers (10:00–10:20) AND 7 customers (10:20–11:00)

First interval (20 min):

$$\mu_1 = \frac{10}{3}$$
$$P(N_1 = 3) = e^{-10/3} \frac{(10/3)^3}{3!}$$

Second interval (40 min):

$$\mu_2 = \frac{20}{3}$$
$$P(N_2 = 7) = e^{-20/3} \frac{(20/3)^7}{7!}$$

Using independence:

$$P = P(N_1 = 3) \cdot P(N_2 = 7)$$
$$P = \left(e^{-10/3} \frac{(10/3)^3}{3!} \right) \left(e^{-20/3} \frac{(20/3)^7}{7!} \right) \approx 0.0102$$

► Poisson Process:

❖ In Conclusion :

- $N(t)$ - counting function for $t \geq 0$
 - This will represent number of events that occurred in $[0, t]$
 - The counting process $\{N(t), t \geq 0\}$, is said to be Poisson process with mean rate λ

► Non-stationary Poisson Process:

❖ Non-stationary Poisson Process

- In many real-world scenarios, **the arrival rate of events is not constant over time.**
- The concept of a non-stationary Poisson process comes into play to model situations where the **event rate changes with time.**
- To model such situations, you can use a non-stationary Poisson process. **In this process, the arrival rate λ is allowed to be a function of time, denoted as $\lambda(t)$.**
- **Example:** More customers in the evening than in the morning.

► Non-Stationary Poisson Process:

- ❖ The key to working a NSPP is the expected number of arrivals by time t , denoted by $\Lambda(t)$. For a time interval $[a, b]$, the mean number of events is:

$$\Lambda(a, b) = \int_a^b \lambda(t) dt$$

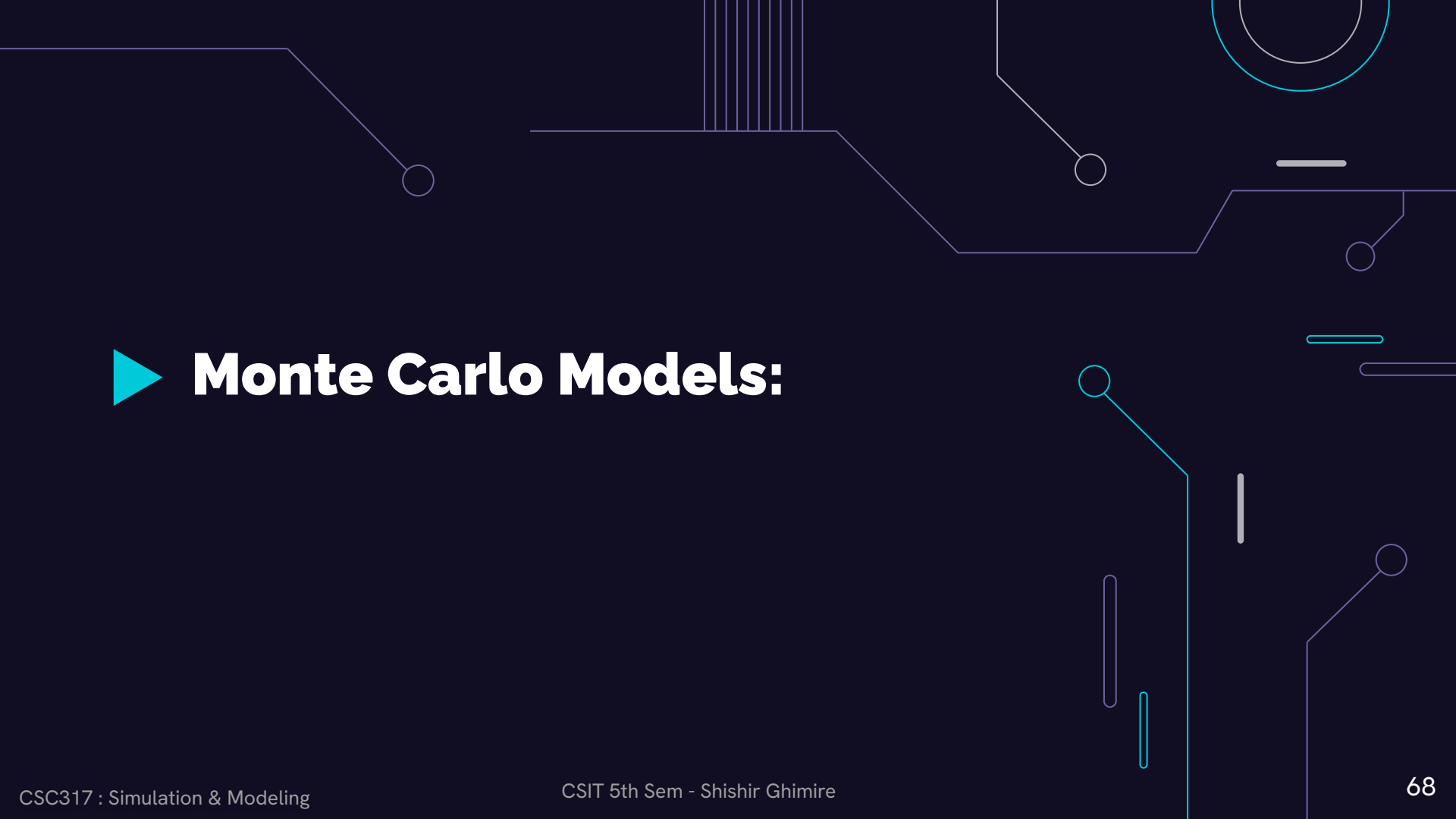
► Batch Arrival Process:

❖ Batch Arrivals:

- In some queueing systems, customers arrive in **groups or “batches,”** such as passengers from a bus. We can represent this using a point process $\Psi = \{t_i\}$, where multiple customers may share the same arrival time:

$$t_1 \leq t_2 \leq t_3 \leq \dots, \text{ and } \lim_{i \rightarrow \infty} t_i = \infty.$$

- Because the sequence extends to infinity, only a finite number of arrival times can be equal.
- Example:
 - **$0 = t_0 = t_1 = t_2 < 1 = t_3 = t_4 = t_5 = t_6 < 3 = t_7 = t_8 \leq t_9 \dots$**
 - This indicates a batch of 3 customers arriving at time 0, a batch of 4 at time 1, and a batch of 2 at time 3, and so on.



► Monte Carlo Models:

► What is **Monte Carlo** ?



► Monte Carlo Method ► History :

- ❖ 1930's: **Enrico Fermi** uses **Monte Carlo** in the calculation of **neutron diffusion**.
- ❖ 1940's: **Stan Ulam** while playing solitaire tries to calculate the **likelihood of winning** based on the initial layout of the cards.
 - After exhaustive combinatorial calculations, he decided to go for **practical approach**.
 - He tries many different layouts and observing the number of successful games.
 - He realized that **computers** could be used to solve such problems.
 - **Stan Ulam** worked with **John Von Neumann** to develop algorithms including **importance sampling and rejection sampling**.
 - **Ulam** and **Von Neumann** suggested that aspects of research into **nuclear fission (building Atom Bomb)** at Los Alamos could be aided by use of computer experiments based on chance.

► Monte Carlo Method ► History :

- ❖ The project was **top secret** so **Von Neumann** chose the name Monte Carlo in reference to the **Casino in Monaco**.
- ❖ **1950's**: Many papers on Monte Carlo simulation appeared in physics literature. The first major MCMC paper was published by Metropolis and Ulam titled "**The Monte Carlo Method**" in 1953.
- ❖ **1970**: Generalization of the **Metropolis algorithm by Hastings** which led to development of MCMC (Markov Chain Monte Carlo).
- ❖ **1980's**: Important MCMC papers appeared in the fields of **computer vision and artificial intelligence** but there were few significant publications in the field of statistics
- ❖ **1990**: MCMC made the first significant impact in statistics in the work of **Gelfand and Smith**.

► Monte Carlo Simulation:

- ❖ The Monte Carlo method is a **numerical method** for **statistical simulation** which utilizes sequences of **random numbers** to perform the simulation.
- ❖ Monte Carlo Simulation is a method used to **predict** and **understand** the behaviour of systems involving **uncertainty**. By running multiple simulations with **random inputs**, this technique helps **estimate** possible outcomes and their probabilities.
- ❖ Instead of relying on a **single solution**, it generates many **different scenarios** to give a clearer picture of the **range** of possible results.
- ❖ Monte Carlo simulations are used to model the **probability of different outcomes** in a process that **cannot easily be predicted** due to the **intervention of random variables**. It is a technique used to understand the impact of risk and uncertainty in prediction and forecasting models.
- ❖ **Underlying concept** is to use randomness, **use repeated random sampling**.

► Probability and Monte Carlo Simulation:

- ❖ The name "Monte Carlo" is inspired by the **Monte Carlo Casino** in Monaco where **randomness** and **chance** are central to the games.
- ❖ Similarly, it uses **random numbers** to predict a **range of possible outcomes**.
- ❖ Three important characteristics of Monte-Carlo method:
 - Its **input distribution** must be **known**.
 - Its **output** must generate **random samples**.
 - Its **result must be known** while performing an experiment.

► What is Monte Carlo Simulation?

- ❖ **Monte Carlo methods** are a widely used class of **computational algorithms** for simulating the behavior of various physical and mathematical systems, and for other computations.
- ❖ **Monte Carlo algorithm** is often a **numerical Monte Carlo method** used to find solutions to mathematical problems (which may have many variables) that cannot easily be solved, (e.g. integral calculus, or other numerical methods).
- ❖ **Monte Carlo methods**, or Monte Carlo experiments, are a broad class of computational algorithms that **rely on repeated random sampling** to obtain numerical results.

► Monte Carlo Simulation ► Applications:

❖ The applications of the Monte Carlo method include:

- Simulation of Natural Phenomena
- Simulation of Experimental Apparatus
- Numerical Analysis
- **Integration:** the integration of a complex, discontinuous, high dimension function is hard to give the analytical result.
- The Simulation of Systems with linear equations
- Sophisticated Statistical Modeling
- Probabilistic Graphical Models
- **Physical Phenomena** - radiation transport in the earth's atmosphere
- **Communications** - bit error rates

► General Algorithm of Monte Carlo Simulation:

- ❖ In general Monte Carlo Simulation is roughly **composed of five steps**:
 1. **Set up probability distributions**: Identify the probability distribution that will be used in the simulation.
 2. Build cumulative probability distributions.
 3. Establish an interval of random numbers for each variable.
 4. **Generate random numbers**: Only accept numbers that satisfy a given condition.
 5. Simulate Trials and Analyze

► General Algorithm of Monte Carlo Simulation:

1. Define Problem and Identify Random Variables

- Understand the system.
- Determine **uncertain** or **random variables** (e.g., demand, lead time, prices).
- Select appropriate **probability distributions** (e.g., normal, uniform, exponential).

2. Build Cumulative Probability Distributions

- **Convert** each probability distribution into a cumulative form.
- This helps in **linking** random numbers to possible outcomes.

► General Algorithm of Monte Carlo Simulation:

3. Create Random Number Intervals

- Divide the **cumulative probability range [0,1]** into **subintervals** corresponding to possible values of the variable.
- Each interval is associated with a specific outcome.

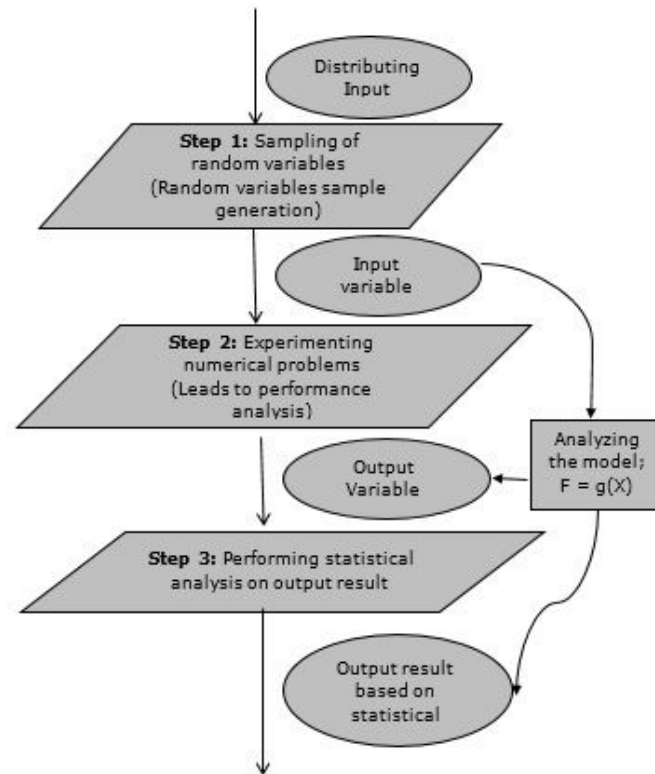
4. Generate Random Numbers

- Use a **random number generator** to generate a sequence of random numbers.
- Assign outcomes based on which interval the number falls into.

5. Simulate Trials and Analyze

- **Repeat the process** for many trials.
- Collect and **analyze** results (mean, variance, etc.) to make data-driven decisions.

► General Algorithm of Monte Carlo Simulation:

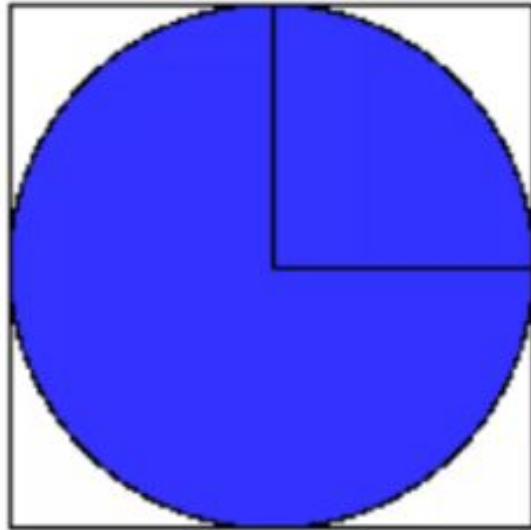


► **Mathematical Statement of Monte Carlo Method:**

Derivation Done in Class

► Monte Carlo Simulation ► Example :

Example 1: Estimating Value of Pi



$$\frac{\# \text{ darts hitting shaded area}}{\# \text{ darts hitting inside square}} = \frac{\text{area of shaded area}}{\text{area of square}}$$

$$\frac{\# \text{ darts hitting shaded area}}{\# \text{ darts hitting inside square}} = \frac{\frac{1}{4} \pi r^2}{r^2} = \frac{1}{4} \pi$$

or

$$\pi = 4 \frac{\# \text{ darts hitting shaded area}}{\# \text{ darts hitting inside square}}$$

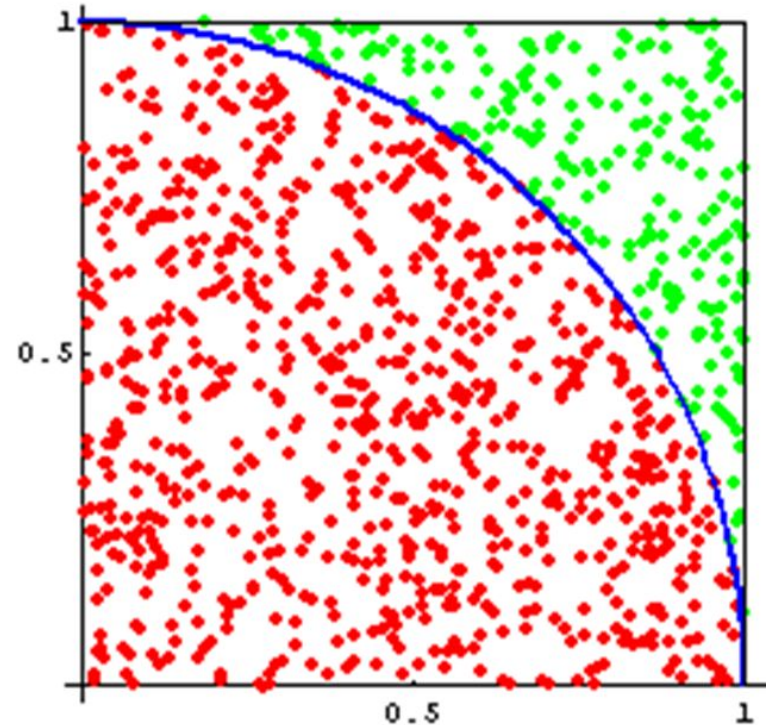
► Monte Carlo Simulation ► Example :

- ❖ Calculation of π : Consider a quadrant (Circular Sector) inscribed in a unit square. Given that the ratio of their areas is $\pi/4$, the value of π can be approximated using a Monte Carlo Method:
 1. Draw a square, then inscribe a quadrant within it.
 2. Uniformly scatter a given number of points over the square.
 3. For each point, we calculate its distance from the origin (0, 0) using the formula: $\text{distance} = \sqrt{x^2 + y^2}$
 4. Count the number of points inside the quadrant (points that satisfy the condition $\sqrt{x^2 + y^2} \leq 1$).
 5. The ratio of the **inside-count** to the **total-sample-count** is an estimate of the ratio of the two areas, $\pi/4$. Multiply the result by 4 to estimate π .

► Monte Carlo Simulation ► π :

Pseudocode to estimate value of π

1. Start
2. Set spoint = cpoint = 0
3. set value of n
4. Loop for range(n)
 Generate x
 Generate y
5. $d = \sqrt{x^2 + y^2}$
6. If $d \leq 1$, then increment cpoint
7. Increment spoint
8. Calculate $\pi = 4 * \text{cpoint} / \text{spoint}$

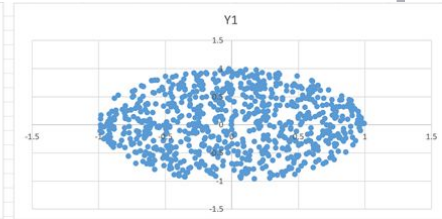
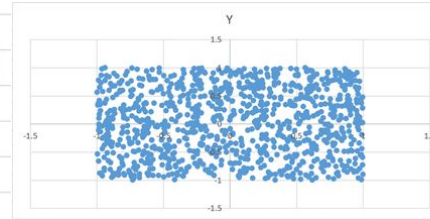


► Calculating Value of π

Calculation Done in Class

► Estimating value of Pi:

bard		Font		Alignment	
		fx		=IF((B3^2+C3^2)<1,1,0)	
B	C	D	E	F	
Points in Rectangle		Points in Circle		Points inside the Circle	
X	Y	X1	Y1	True(1) or False(0)	
0.974007	-0.79348	0	0	0	
0.259482	0.077117	0.259482	0.077117	1	
-0.93821	0.373197	0	0	0	
0.895191	-0.03477	0.895191	-0.03477	1	
-0.57908	-0.7977	-0.57908	-0.7977	1	
0.90583	0.346142	0.90583	0.346142	1	
0.841972	-0.29118	0.841972	-0.29118	1	



=COUNTIF(F2:F1002,"<>0")							
B	C	D	E	F	G	H	I
Points in Rectangle		Points in Circle		Points inside the Circle			
X	Y	X1	Y1	True(1) or False(0)			
0.974007	-0.79348	0	0	0			
0.259482	0.077117	0.259482	0.077117	1		Total Points (N)	1000
-0.93821	0.373197	0	0	0		Points inside Circle (n)	791
0.895191	-0.03477	0.895191	-0.03477	1		n/N	0.791
-0.57908	-0.7977	-0.57908	-0.7977	1		Estimated Pi Value	3.16
0.90583	0.346142	0.90583	0.346142	1			
0.841972	-0.29118	0.841972	-0.29118	1			

► Monte Carlo Simulation ► Applications:

- ❖ **Optimization of Hyperparameters:** It is used to explore various **hyperparameter configurations** in **machine learning models** such as neural networks, to identify the optimal settings for improved model performance.
- ❖ **Reinforcement Learning:** It estimate the value of actions within **reinforcement learning algorithms** helping AI systems learn optimal policies by **simulating and evaluating** different decision-making strategies.
- ❖ **Uncertainty Estimation:** It quantifies the impact of **input uncertainties** on AI model predictions. This helps **improve** the model's robustness and reliability by providing insight into how variations in data affect outcomes.
- ❖ **Risk Assessment in Financial Models:** AI models predicting financial outcomes benefit from it by evaluating the **probability of various financial scenarios**, enabling better risk assessment and decision-making in uncertain environments.
- ❖ **Monte Carlo Tree Search (MCTS):** MCTS is a Monte Carlo-based algorithm is used in **decision-making** for AI in games and other sequential problems, **simulating future states** to guide decision processes and optimize strategies.

► Monte Carlo Simulation ► Advantages :

- ❖ Easy to implement.
- ❖ **Risk Analysis:** It helps to visualize the range of **possible outcomes and assess** the probability of different risks.
- ❖ **Better Decision-Making:** Provides a clearer picture of potential outcomes, providing more **informed decisions**.
- ❖ **Flexibility Across Fields:** Applicable across industries from **finance to engineering to AI**, for various problem types.
- ❖ **Handling Complex Systems:** Models systems with **multiple uncertain variables** that traditional methods **can't handle**.
- ❖ **Visualization of Uncertainty:** Offers a **distribution of outcomes**, making it easier to understand and plan for uncertainty.
- ❖ Can be used for both **stochastic** and **deterministic** problems.

► Monte Carlo Simulation ► Disadvantages :

- ❖ **Time consuming** as there is a need to **generate large number of sampling** to get the desired output.
- ❖ The **results** of this method are only the **approximation of true values**, not the **exact**.
- ❖ **Computational Intensity**: Requires significant time and resources, especially with **complex models** and many simulations.
- ❖ **Quality of Input Data**: The results depend on the **accuracy of the input data**, poor data leads to unreliable outcomes.
- ❖ **Convergence Rate**: High accuracy often needs a **large number of simulations**, especially for rare events.
- ❖ **Over-Simplification**: May miss complex system details or relationships, leading to **potential oversights**.
- ❖ **Interpretation of Results**: Results are **probabilistic** and **misinterpretation** can lead to poor decision-making.

► Scenario for Using Monte Carlo Method Over Other Simulation Methods:

The Monte Carlo Method is **preferred** when:

1. **Complex or Unsolvable Mathematical Models:** Analytical solutions are **difficult** or **impossible** (e.g., multi-dimensional integrals, nonlinear systems).
2. **High Uncertainty in Inputs:** Problems involve **random variables** with **known** probability distributions (finance, risk analysis).
3. **Large-Scale Probabilistic Problems:** Systems with many interacting **random factors** (e.g., weather prediction, nuclear physics).
4. **Statistical Estimation:** When estimating probabilities, expected values, or variances where **direct measurement is impractical**.
5. **Optimization in Uncertain Conditions:** Decision-making under **uncertainty**, especially in operations research and engineering design.
6. **Feasibility of Repeated Random Sampling:** When computational power allows millions of trials to approximate accurate results.

► Stochastic Simulation:

❖ Stochastic Simulation

- Stochastic simulation is a method of **modeling and analyzing systems** whose behavior involves **random variables** or uncertainty.
- It uses **probability distributions** to represent uncertain elements and simulates how the system evolves over time.
- **Example:** Simulating a hospital emergency room to predict **patient wait times**, considering random arrivals and treatment durations.

❖ Applications:

- Queueing systems (banks, call centers).
- Inventory control and supply chains.
- Reliability and risk analysis.
- Manufacturing processes.



How is it different from Monte Carlo?

✓ **Monte Carlo Method** is about using randomness **to solve** numerical problems, while **Stochastic Simulation** is about using randomness **to simulate** real-world systems over time.

► Stochastic Simulation:

◆ Key Features:

- **Randomness Involved:** At least one input or process is probabilistic.
- **Time-dependent:** Often simulates events in chronological order.
- **Multiple Runs:** Repeated simulations (replications) are used to obtain reliable results.
- **Probability Distributions:** Inputs are generated using random numbers and statistical distributions.

◆ Advantages:

- Captures real-world uncertainty.
- Can handle complex systems **without** analytical solutions.
- Provides performance measures like averages, variances, and probabilities.

◆ Limitations:

- Computationally intensive.
- Results are approximate and depend on the quality of the model.
- Requires knowledge of probability and statistics.

► Monte Carlo Method vs Stochastic Simulation

Aspect	Monte Carlo Method	Stochastic Simulation
Definition	A simulation technique that uses random sampling to estimate numerical results for deterministic or probabilistic problems.	A broader simulation technique that models systems influenced by random variables or processes over time.
Purpose	To approximate mathematical results (integrals, probabilities, optimization) using randomness.	To simulate the dynamic behavior of real-world systems that have inherent randomness.
Focus	Primarily focuses on numerical estimation and probability evaluation .	Focuses on modeling system behavior under uncertainty.
System Type	Often used for static problems (no time evolution), but can also be extended to dynamic problems.	Mostly used for dynamic systems that evolve over time (e.g., queues, inventory, population growth).
Time Dependency	Usually time-independent (sampling does not require a time clock).	Time-dependent , simulating events chronologically.
Input	Probability distributions + random number generators.	Probability distributions + event scheduling + system rules.
Output	Estimated numerical value(s) with associated statistical error.	Detailed system performance measures (e.g., waiting times, stock levels, throughput).
Examples	• Estimating the value of π using random points • Risk analysis in finance. • Integration in physics problems.	• Simulating a bank's customer queue. • Modeling traffic flow. • Predicting inventory shortages.

► PYQs:

1. Explain Monte Carlo simulation method with example. 2080, 2079
2. What is feedback system? Explain with example. 2080
3. Why accuracy of analog computer is low? Explain analog computer with suitable example. Differentiate between analog and digital computer. 2079
4. Define arrival pattern. Explain non-stationary Poisson process. 2076
5. Explain the concept of discrete event simulation. Explain poisson arrival pattern. 2078
6. Write short notes on :
 - a. Non-Stationary Poisson process
 - b. Feedback system

► Assignment-2:

Due Date: 16th January, 2026

1. Explain Monte Carlo simulation method with example. 2080, 2079
2. What is feedback system? Explain with example. 2080, 2082
3. Why accuracy of analog computer is low? Explain analog computer with suitable example.
Differentiate between analog and digital computer. 2079
4. Define arrival pattern. Explain Non-Stationary Poisson process. 2076
5. Explain Non Stationary Poisson Process in detail. 2082
6. Explain the concept of discrete event simulation. Explain poisson arrival pattern. 2078
7. Write short notes on :
 - a. Poisson Process
 - b. Stochastic Simulation



▶ **THANKS!**

Do you have any questions?

theciceerguy@p_m.me

9841893035

ghimires.com.np